Day 31 - Midterm Help Session

Cursed Proof of the Euler-Lagrange Equation

We begin with the famous Euler-Lagrange equation:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\right) = \frac{\partial L}{\partial x}$$

Since $\dot{x} = \frac{dx}{dt}$, we rewrite the left-hand side:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial\left(\frac{dx}{dt}\right)}\right) = \frac{\partial L}{\partial x}$$

Now, in a move that would make mathematicians weep, we boldly exchange the differentiation with respect to $\frac{dx}{dt}$ for something that looks like the inverse of $\frac{d}{dt}$. Because obviously:

$$\frac{\partial}{\partial\left(\frac{dx}{dt}\right)} \approx \frac{1}{\frac{d}{dt}} \cdot \frac{\partial}{\partial x}$$

Substitute that into the original equation:

$$\frac{d}{dt}\left(\frac{1}{\frac{d}{dt}} \cdot \frac{\partial L}{\partial x}\right) = \frac{\partial L}{\partial x}$$

Now cancel $\frac{d}{dt}$ with $\frac{1}{\frac{d}{dt}}$, because calculus is just fancy algebra:

$$\frac{\partial L}{\partial x} = \frac{\partial L}{\partial x}$$

Therefore, Q.E.D.

(Q.E.D. = *Questionably Established Derivation*)

Box it, frame it, publish it in *Totally Real Physics Letters*.

Announcements

On Monday

- Homework 8 will be posted (Last HW; Due Nov 21)
- Rubric for final project posted
  - Week 12 - Intro to Lagrangian Dynamics
  - Week 13 - Examples of Lagrangian Dynamics
  - Week 14 - Project Prep (Thanksgiving week)
  - Week 15 - Presentations (Last week of class)
  - Week 16 - Computational Essay Due (Monday of Finals week)

Announcements

Next Week

- Monday and Wednesday: Introduction to Lagrangian Dynamics
- Friday (11/14): DC will be in classroom at 11:30a
    - Hosting speaker @ 12:30p
    - Classroom open from 11:30a-12:50p
    - Second Midterm Help Session